Year 8

REVISION PACK

Unit 1

Name: _____

Class: _____

Unit 1 · Negative numbers

Worked example: calculating with negative numbers

(a) Work out $-12 + (-5)$

(b) Work out -6×9

(c) Work out $84 \div -7$

(d) Sami writes $3 - (-8) = -5$. Is Sami correct? Explain.

(a)	**Step 1**	Plus a minus is minus	$-12 - 5 = -17$
(b)	**Step 1**	Signs different: answer is −	$-6 \times 9 = -54$
(c)	**Step 1**	Signs different: answer is −	$84 \div -7 = -12$
(d)	**Step 1**	Minus a minus is plus	$3 - (-8) = 3 + 8 = 11$
	Step 2	Answer	No, the answer is 11, not -5

Practice: calculating with negative numbers

1.1 (a) Work out $-14 + (-22)$ (b) Work out -6×12 (c) Work out $84 \div -7$ (d) Sami writes $8 - (-13) = -5$. Is Sami correct? Explain.

1.2 (a) Work out $-17 + (-9)$ (b) Work out -8×11 (c) Work out $-72 \div 8$ (d) Sami writes $5 - (-9) = -4$. Is Sami correct? Explain.

1.3 (a) Work out $-25 + (-16)$ (b) Work out 7×-9 (c) Work out $96 \div -12$ (d) Sami writes $3 - (-14) = -11$. Is Sami correct? Explain.

1.4 (a) Work out $-11 + (-33)$ (b) Work out -12×5 (c) Work out $-66 \div -6$ (d) Sami writes $9 - (-4) = 5$. Is Sami correct? Explain.

1.5 (a) Work out $-28 + (-7)$ (b) Work out 13×-4 (c) Work out $132 \div -11$ (d) Sami writes $6 - (-7) = -1$. Is Sami correct? Explain.

1.6 (a) Work out $-19 + (-19)$ (b) Work out -15×6 (c) Work out $-150 \div 10$ (d) Sami writes $10 - (-15) = -5$. Is Sami correct? Explain.

Unit 1 · Prime factors, HCF and LCM

Worked example: prime factors, HCF and LCM

$90 = 2 \times 3^2 \times 5$

(a) Write 84 as a product of prime factors. (b) Work out the HCF of 84 and 90. (c) Work out the LCM of 84 and 90.

(a)	Step 1	Divide by primes	$84 \div 2 = 42,\ 42 \div 2 = 21,\ 21 \div 3 = 7$
	Step 2	Answer	$2^2 \times 3 \times 7$
(b)	Step 1	Primes in both, smaller power	2×3
	Step 2	Answer	6
(c)	Step 1	Every prime, bigger power	$2^2 \times 3^2 \times 5 \times 7$
	Step 2	Answer	1260

Practice: prime factors, HCF and LCM

2.1 $84 = 2^2 \times 3 \times 7$

(a) Write 540 as a product of prime factors. (b) Work out the HCF of 540 and 84. (c) Work out the LCM of 540 and 84.

2.2 $84 = 2^2 \times 3 \times 7$

(a) Write 360 as a product of prime factors. (b) Work out the HCF of 360 and 84. (c) Work out the LCM of 360 and 84.

2.3 $60 = 2^2 \times 3 \times 5$

(a) Write 250 as a product of prime factors. (b) Work out the HCF of 250 and 60. (c) Work out the LCM of 250 and 60.

2.4 $96 = 2^5 \times 3$

(a) Write 144 as a product of prime factors. (b) Work out the HCF of 144 and 96. (c) Work out the LCM of 144 and 96.

2.5 $140 = 2^2 \times 5 \times 7$

(a) Write 420 as a product of prime factors. (b) Work out the HCF of 420 and 140. (c) Work out the LCM of 420 and 140.

2.6 $300 = 2^2 \times 3 \times 5^2$

(a) Write 80 as a product of prime factors. (b) Work out the HCF of 80 and 300. (c) Work out the LCM of 80 and 300.

Unit 1 · Order of operations

Worked example: powers, roots and order of operations

Work out

(a) $5^2 + 6^2 \div 3^2$ (b) $(6^2 + 3^2) \div 3^2$ (c) $2^4 + \sqrt[3]{27} \times 5$ (d) Show that $(-3)^3 = -27$

(a)	Step 1	Powers first	$25 + 36 \div 9$
	Step 2	Then divide	$25 + 4$
	Step 3	Answer	29
(b)	Step 1	Bracket first	$(36 + 9) \div 9$
	Step 2	Then divide	$45 \div 9$
	Step 3	Answer	5
(c)	Step 1	Powers and roots first	$16 + 3 \times 5$
	Step 2	Then multiply	$16 + 15$
	Step 3	Answer	31
(d)	Step 1	Two minuses make plus	$(-3) \times (-3) = 9$
	Step 2	Times by −3 again	$9 \times (-3) = -27$

Practice: powers, roots and order of operations

3.1 Work out

(a) $4^2 + 6^2 \div 2^2$ (b) $(4^2 + 6^2) \div 2^2$ (c) $3^4 + \sqrt[3]{8} \times 5$ (d) Show that $(-2)^3 = -8$

3.2 Work out

(a) $3^2 + 9^2 \div 3^2$ (b) $(3^2 + 9^2) \div 3^2$ (c) $2^4 + \sqrt[3]{125} \times 4$ (d) Show that $(-5)^3 = -125$

3.3 Work out

(a) $8^2 + 4^2 \div 2^2$ (b) $(8^2 + 4^2) \div 2^2$ (c) $3^4 + \sqrt[3]{64} \times 2$ (d) Show that $(-6)^3 = -216$

3.4 Work out

(a) $6^2 + 8^2 \div 2^2$ (b) $(6^2 + 8^2) \div 2^2$ (c) $2^4 + \sqrt[3]{1000} \times 3$ (d) Show that $(-7)^3 = -343$

3.5 Work out

(a) $9^2 + 12^2 \div 3^2$ (b) $(9^2 + 12^2) \div 3^2$ (c) $4^4 + \sqrt[3]{27} \times 10$ (d) Show that $(-1)^3 = -1$

3.6 Work out

(a) $2^2 + 10^2 \div 2^2$ (b) $(2^2 + 10^2) \div 2^2$ (c) $5^4 + \sqrt[3]{216} \times 4$ (d) Show that $(-10)^3 = -1000$

Worked example: multiplying, dividing and powers of powers

Simplify. Leave your answers in index form.

(a) $6^4 \times 6^5$ (b) $6^9 \div 6^2$ (c) $(6^3)^4$

(a)	Step 1	Multiply: add the powers	$4 + 5 = 9$
	Step 2	Answer	6^9
(b)	Step 1	Divide: subtract the powers	$9 - 2 = 7$
	Step 2	Answer	6^7
(c)	Step 1	Bracket: multiply the powers	$3 \times 4 = 12$
	Step 2	Answer	6^{12}

Practice: multiplying, dividing and powers of powers

4.1 Simplify. Leave your answers in index form.

(a) $8^5 \times 8^4$ (b) $8^8 \div 8^2$ (c) $(8^3)^5$

4.2 Simplify. Leave your answers in index form.

(a) $5^4 \times 5^6$ (b) $3^{10} \div 3^4$ (c) $(2^5)^3$

4.3 Simplify. Leave your answers in index form.

(a) $9^2 \times 9^8$ (b) $8^{12} \div 8^5$ (c) $(4^3)^6$

4.4 Simplify. Leave your answers in index form.

(a) $3^5 \times 3^5$ (b) $2^9 \div 2^1$ (c) $(10^4)^2$

4.5 Simplify. Leave your answers in index form.

(a) $4^6 \times 4^3$ (b) $5^{11} \div 5^7$ (c) $(3^2)^8$

4.6 Simplify. Leave your answers in index form.

(a) $2^8 \times 2^7$ (b) $9^6 \div 9^2$ (c) $(5^6)^3$

Unit 1 · Using the laws of indices

Worked example: showing results and writing as a power of 2

(a) Show that $\dfrac{2^5 \times 3}{2^3} = 12$

(b) Show that $\left(\dfrac{3^8}{3^7}\right)^2 = 9$

(c) Write $\dfrac{8 \times 2^4}{32}$ as a power of 2

(d) Write $128 \div 2^3$ as a power of 2

(a)	**Step 1**	Divide: subtract the powers	$2^{5-3} = 2^2$
	Step 2	$2^2 = 4$, then × 3	$4 \times 3 = 12$
(b)	**Step 1**	Inside the bracket	$3^{8-7} = 3^1$
	Step 2	Square it	$3^2 = 9$
(c)	**Step 1**	Write as powers of 2	$8 = 2^3,\ \ 32 = 2^5$
	Step 2	Add, then subtract powers	$3 + 4 - 5 = 2$
	Step 3	Answer	2^2
(d)	**Step 1**	Write as powers of 2	$128 = 2^7$
	Step 2	Subtract the powers	$7 - 3 = 4$
	Step 3	Answer	2^4

Practice: showing results and writing as a power of 2

5.1

(a) Show that $\dfrac{3^3 \times 5}{3^2} = 15$

(b) Show that $\left(\dfrac{5^9}{5^8}\right)^3 = 125$

(c) Write $\dfrac{16 \times 2^5}{32}$ as a power of 2

(d) Write $128 \div 2^3$ as a power of 2

5.2

(a) Show that $\dfrac{2^6 \times 5}{2^4} = 20$

(b) Show that $\left(\dfrac{4^9}{4^8}\right)^3 = 64$

(c) Write $\dfrac{4 \times 2^5}{8}$ as a power of 2

(d) Write $32 \div 2^3$ as a power of 2

5.3

(a) Show that $\dfrac{5^4 \times 3}{5^2} = 75$

(b) Show that $\left(\dfrac{7^5}{7^4}\right)^2 = 49$

(c) Write $\dfrac{32 \times 2^2}{16}$ as a power of 2

(d) Write $128 \div 2^3$ as a power of 2

5.4

(a) Show that $\dfrac{3^6 \times 2}{3^3} = 54$

(b) Show that $\left(\dfrac{2^{10}}{2^7}\right)^2 = 64$

(c) Write $\dfrac{8 \times 2^6}{32}$ as a power of 2

(d) Write $16 \div 2^3$ as a power of 2

5.5

(a) Show that $\dfrac{2^9 \times 9}{2^6} = 72$

(b) Show that $\left(\dfrac{10^8}{10^6}\right)^2 = 10000$

(c) Write $\dfrac{64 \times 2^1}{128}$ as a power of 2

(d) Write $64 \div 2^3$ as a power of 2

5.6

(a) Show that $\dfrac{4^5 \times 5}{4^3} = 80$

(b) Show that $\left(\dfrac{6^{11}}{6^{10}}\right)^3 = 216$

(c) Write $\dfrac{16 \times 2^4}{4}$ as a power of 2

(d) Write $32 \div 2^3$ as a power of 2

Unit 1 · Problems with LCM

Worked example: when things happen together again

Light A flashes every 8 minutes. Light B flashes every 6 minutes. They both flash at 9:00 am. At what time will they next flash together?

Step 1	Multiples of 8	$8, 16, 24, 32$
Step 2	Multiples of 6	$6, 12, 18, 24$
Step 3	First in both: the LCM	24 minutes
Step 4	Add to the start time	$9{:}00 + 24$ min
Step 5	Answer	$9{:}24$ am

Practice: when things happen together again

6.1 Ali's alarm repeats every 28 minutes. Wei's alarm repeats every 35 minutes. Both alarms go off at 6:45 am. When will they next go off together?

6.2 Bus A leaves every 12 minutes. Bus B leaves every 20 minutes. Both leave at 7:00 am. When will they next leave together?

6.3 A red light flashes every 18 minutes. A blue light flashes every 24 minutes. Both flash at 2:00 pm. When will they next flash together?

6.4 Sara swims a length in 4 minutes. Huda swims a length in 6 minutes. They start together at 9:30 am. When are they next at the start together?

6.5 Train A leaves every 25 minutes. Train B leaves every 40 minutes. Both leave at 8:00 am. When will they next leave together?

6.6 One bell rings every 12 minutes. Another bell rings every 16 minutes. They ring together at 10:00 am. When will they next ring together?